

Combined Oxidative Phosphorylation Deficiency 35 (COXPD35) — Comprehensive Disease Report

MONDO: MONDO:0054742 · **OMIM:** #617873 · **Gene:** *TRIT1* (tRNA isopentenyltransferase 1)
Category: Mendelian, autosomal-recessive mitochondrial (nuclear-encoded) disorder **Report date:** 2026-10-01 · **Evidence base:** Human clinical case reports/series; in vitro (patient fibroblasts, HeLa); model organism (yeast). ~15–20 patients reported worldwide.

Scope note. COXPD35 is an ultra-rare disease documented almost exclusively through single-patient/single-family case reports. Quantitative epidemiology, survival statistics, QoL metrics, and controlled trials **do not exist**. Where a claim rests on general mitochondrial-disease knowledge rather than *TRIT1*-specific data, this is stated explicitly. Primary citations use PMIDs.

1. Disease Information

Overview. COXPD35 is a rare autosomal-recessive mitochondrial disease caused by biallelic pathogenic variants in *TRIT1*, which encodes tRNA isopentenyltransferase 1 (IPTase). The enzyme adds an isopentenyl group to adenosine-37 (i6A37) in the anticodon loop of a small subset of **both cytosolic and mitochondrial tRNAs**; its loss produces tRNA hypomodification, defective mitochondrial (and cytosolic) protein synthesis, and a combined deficiency of oxidative-phosphorylation (respiratory-chain) complexes. Clinically it is a neurodevelopmental mitochondrial encephalopathy dominated by microcephaly, developmental delay/intellectual disability, and epilepsy

([P 24901367](#);
[P 28185376](#)).

Key identifiers. - OMIM: #617873 (Combined oxidative phosphorylation deficiency 35) - MONDO: MONDO:0054742 (cross-refs confirmed via Monarch: OMIM:617873, DOID:0111464, UMLS:C4693466, MedGen C1639653) - Gene: *TRIT1*, HGNC:20286, NCBI Gene ID 54802, Ensembl ENSG00000043514, UniProt Q9H3H1, chromosome 1p34.2 - Orphanet: **no COXPD35-specific ORPHA code** (absent from MONDO:0054742 mappings); grouped clinically under "Combined oxidative phosphorylation defect"/non-syndromic mitochondrial encephalopathy - ICD-10: best fit E88.49 (other mitochondrial metabolism disorders); ICD-11: 5C53.1 (mitochondrial metabolism disorders). No disease-specific code. - MeSH: no specific descriptor; indexed under "Mitochondrial Diseases" / "OXPHOS" terms.

Synonyms / alternative names. COXPD35; Combined oxidative phosphorylation deficiency 35; TRIT1 deficiency; TRIT1-related mitochondrial disease; tRNA isopentenyltransferase 1 deficiency.

Information source. Disease-level knowledge is aggregated from **individual-patient case reports** (EHR-derived descriptions published as case studies) plus functional/biochemical studies in patient cells and model organisms. No registry or EHR-cohort data exist.

2. Etiology

Primary cause (genetic). Biallelic (homozygous or compound-heterozygous) loss-of-function variants in *TRIT1*. The disorder is monogenic and Mendelian; there is no environmental or infectious cause. The first case was a homozygous **p.Arg323Gln** identified by exome sequencing in a patient with "severe combined mitochondrial respiratory chain defects and corresponding perturbation in mitochondrial protein synthesis" (

P 24901367

Causality was consolidated by matchmaking across additional recessive families (

P 28185376

Genetic risk factors. The causal genotype is biallelic *TRIT1* LOF. **Consanguinity** increases homozygosity risk (several reported families are consanguineous, e.g., Palestinian case,

P 41760017

general consanguinity/IEM link,
 P 37453291

No established susceptibility loci or validated modifier genes.

Environmental / protective factors. None established. As a fully penetrant recessive metabolic disease, environmental exposures are not primary drivers. (General mitochondrial-disease principle: mitochondrial toxins, fasting/catabolic stress, fever, and certain drugs—e.g., valproate, aminoglycosides—can exacerbate decompensation; **inferred**, not demonstrated for TRIT1.)

Gene–environment interactions. Not characterized. Plausibly, intercurrent infection/metabolic stress precipitates seizures and decompensation (as in the Palestinian patient who presented in status epilepticus with concurrent pneumonia,

P 41760017
 — **inferred**.

3. Phenotypes

Phenotypes compiled from reported cases
 (P 24901367

28185376; 32948376; 35418828; 36049610; 41760017). Frequencies are qualitative given the tiny N.

Phenotype	Type	HPO suggestion	Onset	Severity/ course	Frequency
Microcephaly (often progressive)	Physical/sign	HP:0000252 (microcephaly); HP:0005484 (postnatal)	Congenital/ infantile	Variable, often progressive	Very common
Global developmental delay / intellectual disability	Behavioral/ cognitive	HP:0001263; HP:0001249	Infantile	Mild→severe; often severe	Very common
Epilepsy / seizures (incl. pharmaco- resistant, status epilepticus)	Sign/neurologic	HP:0001250; HP:0002133 (status epilepticus)	Early- infantile	Episodic, frequently drug- resistant	Common/ characteristic
Abnormal EEG	Lab/ electrophysiology	HP:0002353	Infantile	—	Reported
Thin/ hypoplastic corpus callosum	Imaging/sign	HP:0033725 / HP:0002079	Congenital	Static structural	Reported
Dysmorphic facial features	Physical	HP:0001999	Congenital	—	Reported
Hypotonia / motor impairment	Sign	HP:0001252	Infantile	Variable	Reported
Failure to thrive / feeding difficulty	Sign	HP:0001508	Infantile	—	Reported
	Lab abnormality	HP:0002151 (hyperlactatemia)	—	Variable/ inconsistent	Variable

Phenotype	Type	HPO suggestion	Onset	Severity/ course	Frequency
Elevated lactate (serum/CSF)					
Decreased mitochondrial complex I activity	Lab	HP:0011923	—	—	In biochemically tested cases
Decreased mitochondrial complex IV activity	Lab	HP:0008347	—	—	In biochemically tested cases

Additional HPO-curated features (Monarch annotation set for MONDO:0054742, derived from reported cases): - **Seizure semiology:** epileptic encephalopathy (HP:0200134), generalized myoclonic seizure (HP:0002123), febrile seizure (HP:0002373). - **Motor / movement:** generalized hypotonia (HP:0001290), spasticity (HP:0001257), dystonia (HP:0001332), delayed ability to walk (HP:0031936). - **Speech / cognition:** absent speech (HP:0001344), delayed speech and language development (HP:0000750). - **Neuroimaging:** cerebral atrophy (HP:0002059) (in addition to thin corpus callosum). - **Ophthalmologic:** optic disc hypoplasia (HP:0007766), myopia (HP:0000545), esotropia (HP:0000565).

Characteristics. Onset is typically **neonatal–early-infantile**. Severity is **variable**: the literature explicitly documents "both very severe, with generalized pharmaco-resistant seizures, and mild phenotypes" (

P 36049610

Course is generally **progressive/static-encephalopathic** with episodic seizures.

QoL impact. Severe forms: profound disability, intractable epilepsy, dependence for all care, risk of early death (**data not quantified**; no EQ-5D/SF-36 data). Milder forms retain more function.

4. Genetic / Molecular Information

Causal gene. *TRIT1* (HGNC:20286; OMIM gene 606434; UniProt **Q9H3H1**), chromosome **1p34.2**. Encodes **tRNA dimethylallyltransferase / tRNA isopentenyltransferase 1 (IPTase)**, a **467-aa**, dual-targeted (**mitochondrial + cytoplasmic**, confirmed UniProt subcellular annotation) enzyme, **EC 2.5.1.75**, catalyzing: adenosine(37) in tRNA + dimethylallyl diphosphate → N6-dimethylallyladenosine(37) in tRNA + diphosphate. Domain architecture: **Pfam PF01715 (IPPT)**; InterPro **IPR027417 (P-loop NTPase fold)** and **IPR003604 (zinc-finger)** — the latter is disrupted by the p.Glu327Lys variant

([P 32948376](#)).

UniProt annotates **6 isoforms**, the molecular basis for the alternative-splicing buffering discussed below

([P 15870694](#)).

Representative pathogenic variants (all germline, biallelic recessive). - **c.968G>A, p.Arg323Gln** — homozygous, first reported; missense near the tRNA-binding surface (modeled on yeast IPTase co-crystal), severe combined RC defect

([P 24901367](#)).

- **c.979G>A, p.Glu327Lys + c.682+2T>C (splice)** — compound het, Korean siblings; p.Glu327Lys disrupts the zinc-finger/IPT conformation

([P 32948376](#)).

- **Novel missense variant — COXPD35 case**

([P 35418828](#)).

- **Four novel variants in two patients** (missense/splice/truncating) expanding the allelic and phenotypic spectrum

([P 36049610](#)).

- **Two novel compound-heterozygous variants — first Palestinian case**

([P 41760017](#)).

Variant classes: missense, splice-site (e.g., c.682+2T>C), and truncating (nonsense/frameshift) — all converging on loss of IPTase function. **Classification:** reported variants are pathogenic/likely pathogenic by ACMG (segregation, functional assays, matchmaking;

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ClinVar: 208 *TRIT1* records, of which **67 are pathogenic/likely-pathogenic** (accessed 2026; NCBI E-utilities). **Functional consequence:** loss of function (reduced/abolished i6A37 modification), confirmed by complementation

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Population-genetic constraint (gnomAD v4; this analysis). Ensembl ENSG00000043514, chr1:39,838,110–39,883,511 (GRCh38; = 1p34.2). *TRIT1* is **LoF-tolerant: pLI $\approx$ 0** (6.4×10^{-13}), **observed/expected LoF (LOEUF) = 0.84** (90% CI 0.66–1.08; 43 observed vs 51.2 expected LoF), **missense Z = 0.89**. Individual pathogenic alleles are rare, but some predicted-LoF alleles are relatively common (e.g., p.Arg112GlufsTer36 AF = 7.6×10^{-4} ; p.Arg327Ter AF = 6.4×10^{-4}). **Interpretation:** heterozygous LoF is tolerated (carriers healthy; recessive inheritance), and because several predicted-LoF alleles are common yet the disease is ultra-rare, many such alleles are likely **not fully penetrant** — consistent with *TRIT1*'s complex alternative splicing in which **only the full-length isoform is catalytically active**

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allowing residual isoform activity to buffer some truncations. Disease therefore likely requires biallelic variants that abolish activity across functional isoforms.

Modifier genes / epigenetics / chromosomal abnormalities. None established for COXPD35. (Separately, *TRIT1* promoter methylation and expression are studied in cancer contexts —

P 40391218

— but this is unrelated to COXPD35.)

5. Environmental Information

No environmental, lifestyle, or infectious etiologic factors. COXPD35 is purely genetic. Intercurrent illness/metabolic stress may act as a **trigger for decompensation** (inferred;

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describes presentation with concurrent pneumonia). Standard mitochondrial-disease caution regarding mitotoxic drugs applies but is not *TRIT1*-specific.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic LOF variants in *TRIT1*** → loss/reduction of tRNA isopentenyltransferase (IPTase) enzymatic activity (demonstrated;

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 2. Reduced IPTase activity **results in i6A37 hypomodification** of the A36-A37-A38 anticodon-loop subset of tRNAs in **both cytosol and mitochondria** (demonstrated in patient cells;

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 substrate set defined in

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 3. Affected substrates: cytosolic **tRNA-Ser(AGA/CGA/UGA)** and **selenocysteine tRNA[Ser]Sec(UCA)**; mitochondrial **mt-tRNA-Ser(UCN)** and **mt-tRNA-Trp(UCA)**
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 4. i6A37 loss **leads to** decreased **translational efficiency and fidelity** in a codon-specific manner (yeast-established principle;

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 5. Impaired **mitochondrial protein synthesis leads to** reduced levels of mtDNA-encoded OXPHOS subunits → **combined respiratory-chain (complex I/III/IV) deficiency** (decreased select mitochondrial proteins shown;

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 6. **Branch (cytosolic, demonstrated in model):** i6A37 loss also impairs **cytosolic translation** of specific codon-biased mRNAs. In *S. pombe*, the "mitochondrial-like" ATP/energy deficit is **fully rescued by a cytosol-only IPTase** and by cytosolic tRNA overexpression — cy-tRNA-Tyr is rate-limiting, and Tyr-codon-rich mRNAs encode carbon-metabolizing (often mitochondrially localized) enzymes
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- This shows the OXPHOS/energy phenotype is **not exclusively** due to mt-tRNA

hypomodification; a cytosolic-translation branch contributes (human cytosolic substrates differ — tRNA-Ser/SeC).

7. **Branch (selenoprotein):** reduced i6A37 on tRNA[Ser]Sec may **impair selenoprotein translation** (UGA recoding), potentially compromising antioxidant defense — **inferred**

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P 24126054
 shows tRNA[Ser]Sec is a partial substrate; selenoprotein/tRNA[Ser]Sec interplay studied in
P 34768885).

8. Deficient OXPHOS → **ATP-synthesis failure and secondary oxidative stress** → energy crisis in metabolically demanding **neurons** → **impaired brain growth (microcephaly), neurodevelopmental delay, and neuronal hyperexcitability (epilepsy)** (clinical phenotype;

P 28185376).

9. **Clinical manifestation:** mitochondrial encephalopathy with variable severity set by residual TRIT1 activity.

Checklist coverage

- **Molecular pathway:** tRNA wobble/anticodon modification → mitochondrial translation → OXPHOS/electron transport chain (KEGG oxidative phosphorylation; Reactome mitochondrial translation).
- **Cellular processes:** mitochondrial translation (GO:0032543), tRNA isopentenylation / tRNA wobble modification (GO:0006400 tRNA modification; GO:0009972 cytokinin biosynthesis-related transferase activity not applicable), bioenergetic failure, likely neuronal apoptosis (inferred).
- **Protein dysfunction:** loss of function; missense variants destabilize the catalytic/zinc-finger and tRNA-binding regions

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 32948376).

- **Metabolic changes:** reduced ATP, lactate elevation (variable), combined complex deficiency.
- **Biochemical abnormality:** deficiency of i6A37 modification (enzyme-substrate defect); EC 2.5.1.75 (tRNA dimethylallyltransferase).

- **Immune involvement:** none (not an immunologic disease).
- **Tissue-damage mechanism:** chronic energy deficit + oxidative stress in CNS.

Upstream vs downstream: *TRIT1* LOF and tRNA hypomodification are **upstream**; OXPHOS deficiency, energy failure, and neurologic manifestations are **downstream**. **Cell types:** neurons (CL:0000540), potentially skeletal myocytes (CL:0000188), fibroblasts used as model. **GO terms:** GO:0006400 (tRNA modification), GO:0032543 (mitochondrial translation), GO:0006119 (oxidative phosphorylation). **Subcellular:** mitochondrion (GO:0005739), cytosol (GO:0005829).

7. Anatomical Structures Affected

- **Organ level (primary):** brain/CNS (UBERON:0000955; brain). Secondary: skeletal muscle (UBERON:0001134) in some mitochondrial presentations. **Body system:** nervous system (predominant).
- **Imaging-defined sites:** cerebral cortex (microcephaly), **corpus callosum** (thinning/hypoplasia; UBERON:0002336).
- **Tissue/cell level:** neurons (CL:0000540); patient dermal fibroblasts (CL:0000057) used experimentally.
- **Subcellular: mitochondrion** (GO:0005739; UBERON cell component) — primary; **cytosol** (GO:0005829) for cytosolic tRNA modification.
- **Lateralization:** bilateral/diffuse CNS involvement.

8. Temporal Development

- **Onset:** congenital to early-infantile; seizures and delay typically manifest in the **first months of life**

(P 32948376).

- **Pattern:** chronic, static-to-progressive encephalopathy with **episodic seizures**; metabolic decompensation can be acute (status epilepticus;

P 41760017).

- **Progression rate:** variable — severe lethal infantile forms vs milder slowly-progressive forms

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- **Duration:** lifelong.
- **Critical period:** early neurodevelopment (first years) is the window of greatest vulnerability and of potential supportive-intervention benefit (inferred;

P 37453291
 emphasizes early recognition in neurometabolic epilepsies).

- **Remission:** none; seizures may be partially controlled but not cured.

9. Inheritance and Population

- **Inheritance:** autosomal recessive

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- **Penetrance:** effectively complete for biallelic LOF; **expressivity variable**

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- **Anticipation / germline mosaicism:** not applicable/not reported.
- **Consanguinity / founder effects:** consanguinity contributes (homozygous cases;

P 41760017);
 no defined founder allele.

- **Carrier frequency:** heterozygous carriers asymptomatic (gnomAD: *TRIT1* is LoF-tolerant, pLI≈0, LOEUF 0.84; this analysis). A naive cumulative predicted-LoF allele frequency of $\sim 2.1 \times 10^{-3}$ would imply a carrier rate of $\sim 1/236$ and recessive incidence $\sim 1/223,000$ — but this **markedly overestimates** true disease frequency because many predicted-LoF alleles appear non-penetrant (see §4), so the real incidence is far lower.
- **Epidemiology: ultra-rare** — only ~a dozen+ patients reported worldwide ("Only 10 patients have been reported,"

P 36049610;
 "Only six cases ... reported worldwide" earlier;

P 32948376

Prevalence/incidence **not quantifiable** but observed frequency is far below the naive gnomAD prediction, indicating reduced penetrance of many LoF alleles; likely $<<1/1,000,000$ for the clinically recognized disease.

- **Demographics:** reported across diverse populations (European, Korean —

P 32948376

Turkish —

P 35418828

Palestinian —

P 41760017

Sex ratio: no sex bias expected (autosomal); not formally reported. **Age distribution:** pediatric-onset.

10. Diagnostics

- **Primary diagnostic route: genetic testing** — whole-exome sequencing (WES) is the workhorse that established the diagnosis in essentially all reported cases, confirmed by **Sanger sequencing**

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32948376; 28185376). WGS, mitochondrial/nuclear gene panels, and single-gene *TRIT1* testing are alternatives. Chromosomal microarray/karyotype/FISH are generally normal (point-variant disease).

- **Biochemical/clinical tests:** respiratory-chain enzyme assays in muscle/fibroblasts show **combined complex deficiency**; serum/CSF **lactate** may be elevated (variable). Research-grade assay: **i6A37 quantification** in tRNA (mass spec/HPLC) is diagnostic/functional confirmation

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24126054).

- **Functional confirmation:** yeast complementation and TRIT1-rescue of patient fibroblasts validate variant pathogenicity

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15870694). A **tRNA-modification-sensitive northern blot** (differential probe annealing to modified vs unmodified tRNA) can quantify i6A37 hypomodification on specific tRNAs and has been used to characterize TRIT1 and other tRNA-modification-enzyme deficiencies

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P 34930808).

- **Imaging:** brain MRI — microcephaly, thin corpus callosum, non-specific changes

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P 41760017).

EEG — epileptiform/abnormal background.

- **Differential diagnosis:** other combined-OXPHOS deficiencies and mitochondrial aminoacyl-tRNA synthetase / tRNA-modification disorders (e.g., MTO1, GTPBP3, TRMU, MARS2, other COXPD subtypes), early-infantile epileptic encephalopathies, and other causes of microcephaly + epilepsy + developmental delay. Distinguished by gene identification.
- **Screening:** no newborn screening; **cascade/carrier testing** of relatives and **prenatal/preimplantation testing** feasible once familial variants are known.

11. Outcome / Prognosis

- **Survival/mortality:** no cohort statistics. Severe forms carry high morbidity and risk of early childhood death; milder forms survive with disability

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Disease-specific mortality **not quantified**.

- **Morbidity/function:** profound neurodevelopmental disability, drug-resistant epilepsy, dependence for care in severe cases.
- **Complications:** status epilepticus, aspiration/respiratory infections (e.g., pneumonia;

P 41760017),

feeding failure, metabolic decompensation.

- **Prognostic factors:** **residual TRIT1/i6A37 activity** (genotype–phenotype correlation: truncating/severe LOF → worse) appears to drive severity

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earlier/more severe seizures portend worse outcome (general neurometabolic principle,
 (P 37453291)).

- **QoL measures:** none published.

12. Treatment

No disease-specific or curative therapy exists. Management is **supportive and multidisciplinary** (general mitochondrial-disease standard;
 (P 30024619)).

- **Seizure control / pharmacotherapy:** standard antiepileptic drugs; many patients are **pharmaco-resistant**

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Acute status epilepticus managed with benzodiazepines + phenytoin and supportive care

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 (P 41760017)

Caution: avoid valproate where mitochondrial dysfunction suspected (general principle).

- **Metabolic/"mitochondrial cocktail" (NCIT: Coenzyme Q10 therapy, etc.):** antioxidants and enzyme cofactors — **Coenzyme Q10/ubiquinone, riboflavin (vitamin B2), creatine monohydrate, α-lipoic acid, L-carnitine, arginine/citrulline, folinic acid, vitamins C/E/K** — are used empirically; evidence is low-quality and non-specific to TRIT1

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 (P 30024619)

NCIT terms: Coenzyme Q10, Riboflavin, Levocarnitine, Thiamine.

- **Nutritional support:** individualized caloric/nutrient support; avoid fasting/catabolic stress

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- **Supportive/rehabilitative:** physical, occupational, speech therapy; feeding support (NG/gastrostomy); respiratory care; developmental support.
- **Advanced/experimental therapeutics: none approved;** no gene, cell, or RNA therapy trials for COXPD35. Gene replacement is a theoretical future avenue (complementation already restores the biochemical defect in vitro —

P 24901367

but **not clinical**.

- **Pharmacogenomics / targeted / immunotherapy:** not applicable.

13. Prevention

- **Primary prevention: genetic counseling** for at-risk/consanguineous families; carrier testing; **prenatal diagnosis and preimplantation genetic testing** once familial *TRIT1* variants are known. No vaccine/behavioral prevention (non-infectious, non-lifestyle disease).
- **Secondary prevention:** early genetic diagnosis via WES enables timely supportive care and seizure management

(P 37453291

stresses early recognition in neurometabolic epilepsies).

- **Tertiary prevention:** aggressive seizure control, infection prophylaxis, nutrition, and avoidance of mitotoxic drugs/catabolic stress to limit complications (inferred/general).
- **Public-health/environmental interventions:** not applicable beyond consanguinity counseling in high-prevalence communities.

14. Other Species / Natural Disease

- **Taxonomy / orthologs (NCBI Taxon; evolutionary conservation):** IPTase is deeply conserved — *E. coli* **MiaA**, *S. cerevisiae* **Mod5**, *S. pombe* **Tit1**, *C. elegans* **GRO-1**, human **TRIT1**

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This conservation underpins cross-species functional assays.

- **Natural disease in animals:** no naturally occurring TRIT1 disease reported in companion animals/wildlife (OMIA — *not documented*). Veterinary relevance: none established.

- **Comparative biology:** substrate repertoire is **narrower in humans** (4 cytosolic tRNA-Ser/SeC substrates) than in yeasts, which modify a broader tRNA set

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— relevant when extrapolating yeast phenotypes.

- **Transmission/zoonosis:** not applicable (genetic disease).

15. Model Organisms

- **Yeast (primary functional models):** *S. cerevisiae* **Mod5** and *S. pombe* **Tit1** are used for **complementation assays** to test human *TRIT1* variant function and nonsense-suppressor activity

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P 15870694);

24901367). Yeast established that i6A37 loss reduces translational efficiency/fidelity codon-specifically. The *S. pombe* **tit1-Δ** deletion is a functional disease model reproducing a "**mitochondrial-like**" **metabolic deficiency** (reduced ATP, impaired growth on glycerol/under rapamycin) that is dissectible to a cytosolic-tRNA mechanism

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P 26857223).

- **Bacterial/invertebrate orthologs:** *E. coli* MiaA; *C. elegans* GRO-1 (clk-type/developmental-timing phenotypes) — comparative references

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- **Human cell models:** patient-derived **fibroblasts** (i6A37 deficiency; rescued by WT-TRIT1 transduction —

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and **HeLa** siRNA-knockdown (substrate mapping —

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- **Phenotype recapitulation:** yeast/cell models faithfully reproduce the **biochemical** defect (i6A37 loss, translation perturbation) and allow genotype–function correlation, but **do not** reproduce the human neurodevelopmental/epilepsy phenotype. A dedicated vertebrate (mouse/zebrafish) COXPD35 disease model is **not reported** — a search of the Alliance of

Genome Resources returned no characterized mouse *Trit1* disease allele/model (accessed 2026), highlighting a key resource gap. *C. elegans* **gro-1** (IPTase ortholog) mutants show developmental-timing/clock phenotypes and provide comparative insight into i6A37 biology.

- **Applications:** variant pathogenicity classification, mechanism of translation defect, substrate identification, and complementation proof-of-concept for gene replacement.

Supported vs. Refuted Hypotheses

Supported: - COXPD35 is caused by biallelic LOF in *TRIT1* (P 24901367 28185376). - Core phenotype = microcephaly + developmental delay/ID + epilepsy, with a severity spectrum (P 28185376 36049610). - Mechanism = i6A37 tRNA hypomodification impairing mitochondrial (and cytosolic) translation → combined OXPHOS deficiency (P 24901367 24126054; 28185376). - Autosomal-recessive, ultra-rare; diagnosis by WES + Sanger (P 32948376 41760017).

Refuted / not applicable: - Environmental, infectious, or lifestyle etiology (none). - Existence of curative or disease-specific pharmacotherapy (none; supportive only). - Dominant/X-linked/mitochondrial-DNA inheritance (it is nuclear autosomal-recessive).

Limitations & Future Directions

- Evidence rests on <~20 patients; no epidemiologic, survival, or QoL data; genotype–phenotype correlation is suggestive but statistically underpowered.

- No vertebrate disease model; no trials. Future needs: a knock-in mouse/zebrafish model, systematic i6A37 biomarker assays, natural-history registry, and exploration of selenoprotein/antioxidant contributions and AAV gene-replacement given in vitro rescue.

Key Ontology Terms

- **MONDO:** MONDO:0054742 • **Gene/HGNC:** HGNC:20286 (*TRIT1*)
- **HPO (core):** HP:0000252 (microcephaly), HP:0001263 (global developmental delay), HP:0001249 (intellectual disability), HP:0001250 (seizures), HP:0002133 (status epilepticus), HP:0200134 (epileptic encephalopathy), HP:0002123 (generalized myoclonic seizure), HP:0002079 (hypoplasia of corpus callosum), HP:0002059 (cerebral atrophy), HP:0002151 (hyperlactatemia), HP:0011923 (decreased complex I), HP:0008347 (decreased complex IV)
- **HPO (additional):** HP:0001290 (hypotonia), HP:0001257 (spasticity), HP:0001332 (dystonia), HP:0001344 (absent speech), HP:0007766 (optic disc hypoplasia), HP:0000545 (myopia), HP:0000565 (esotropia), HP:0001508 (failure to thrive)
- **GO:** GO:0006400 (tRNA modification), GO:0032543 (mitochondrial translation), GO:0006119 (oxidative phosphorylation); GO:0005739 (mitochondrion), GO:0005829 (cytosol)
- **CL:** CL:0000540 (neuron), CL:0000057 (fibroblast)
- **UBERON:** UBERON:0000955 (brain), UBERON:0002336 (corpus callosum), UBERON:0001134 (skeletal muscle)
- **CHEBI:** CHEBI:dimethylallyl-diphosphate (DMAPP, isopentenyl donor); i6A (N6-isopentenyladenosine)
- **NCIT (treatment):** Coenzyme Q10, Riboflavin, Levocarnitine (supportive/empiric)

Primary References (PMID)

24901367 · 28185376 · 24126054 · 26857223 · 34930808 · 34768885 · 15870694 · 32948376 · 35418828 · 36049610 · 41760017 · 30024619 · 37453291 · 40391218